ROADWATCH

Geospatial road-defect intelligence for municipal maintenance in Chennai

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DESIGN OF SMART CITIES

Tamil Nadu Smart Cities Project

MAP • TRIAGE • SPATIAL ANALYTICS

30 structured records in current source	**4** severity levels	**5** workflow states	**3** time windows: 24h / 7d / 30d

WHY ROADWATCH?

Municipal road teams need to connect a defect's place, severity and repair state. A map gives geography; a table gives record detail. Roadwatch brings both into one review workflow.

KEY CONTRIBUTION

- Shared map + table selection
- Fast filtering and record triage
- Summary KPIs for operational review

WHAT IS IMPLEMENTED

MAP-FIRST VIEW
Leaflet street basemap with severity-coded circle markers and pop-ups

LINKED RECORDS
Marker and table row open a shared detail view

OPERATIONS FILTERS
Search, sorting, severity/status chips and 24h / 7d / 30d ranges

SUMMARY ANALYTICS
KPIs, severity mix, detection trend and mean record confidence

WORKFLOW STATES

NEW | IN REVIEW | SCHEDULED | REPAIRED | DISMISSED

Browser-side record state; centralized persistence is a next integration.

DATA MODEL & OPERATIONAL LABELS

Each record can carry: ID • road / area • latitude & longitude • severity • confidence • detection time • status • assigned crew • depth / width.

SEVERITY

MINOR | MODERATE | SEVERE | CRITICAL

WORKFLOW STATUS

New | In review | Scheduled | Repaired | Dismissed

DATA QUALITY RULE: require a coordinate source and timestamp; preserve confidence provenance; record depth / width units; flag missing or implausible values for review.

CURRENT BOUNDARY & DATA SAFEGUARDS

PRESENT IN THE BUILD

- Responsive HTML/CSS interface
- JavaScript record + filter state
- Leaflet + CARTO street basemap
- OSM map attribution
- Search, sort, KPI and detail drawer
- Severity / status actions

NOT YET CONNECTED

- Embedded vision inference
- Live camera / GPS ingestion
- REST API + database persistence
- Authentication + audit trail
- Verified field measurements
- Measured model accuracy

BEFORE OPERATIONAL USE

Check points against the road network and municipal boundary; store source + time for each report; review uncertain records rather than auto-dispatch; avoid claims about detection accuracy until holdout testing is complete.

TRUSTWORTHY DATA → TRUSTWORTHY WORK ORDERS

Keep a stable record ID, coordinate precision, time, severity source, status-change history and responsible crew. Duplicate handling and long-term auditability depend on server-side persistence.

SYSTEM ARCHITECTURE

RECORD SOURCE (Current JS data; future API) › FILTER STATE (Time, severity, status, search) › VIEWS (Map, table, KPIs, trends) › TRIAGE (Review, schedule, repair) › PERSISTENCE (Browser state now; DB next)

A single filtered record set drives the map, table and summary panels so an operator can move from spatial context to record-level triage without losing selection context.

DATA-SCIENCE EXTENSION

1	CAPTURE	Road image/video + GPS + timestamp
2	ANNOTATE	Label potholes / cracks; quality review
3	DETECT	Train baseline object detector; confidence output
4	GEO-ENRICH	Attach position; snap to roads; group duplicates
5	ANALYZE	Hotspots, risk queue, repeat reports and repair trends

Integration status: dashboard layer is implemented; detector training, live ingestion and automatic geospatial grouping remain future work.

FROM COORDINATES TO ACTIONABLE INSIGHT

HOTSPOT RATE	Unique defects per road-kilometre per time window. Normalizes counts by segment length.
CANDIDATE PRIORITY	Score = S × C × E. S = severity weight (1–4), C = detector confidence (0–1), E = exposure factor (1–3). Tune with municipal staff.
REPEAT REPORTS	Group reports when geographic distance ≤ r and time gap ≤ τ. Choose r and τ from GPS error and local reporting cadence.
MAINTENANCE PERFORMANCE	After durable timestamps are available: backlog by zone / crew, median detection-to-triage time and median time-to-repair.

These are proposed analytics, not features currently reported by the interface.

DATASET & REPRODUCIBLE EVALUATION

47,420	55k+	6	4
images	damage instances	countries	damage classes

RDD2022 is an external benchmark described in the literature, with images from India and five other countries. Its labelled road damage includes longitudinal cracks, transverse cracks, alligator cracks and potholes. It is a candidate data source; it is not connected to the current dashboard.

SUGGESTED EXPERIMENT PROTOCOL

1 Split by route / recording session, not adjacent frames.
2 Report class counts and imbalance before training.
3 Compare a baseline detector with a tuned candidate.
4 Validate on a held-out set; then test Chennai-specific roads, daylight, rain, blur and camera angles.
5 Freeze thresholds before reporting final scores.

RDD2022 paper

Local calibration is essential: benchmark performance may not transfer to Chennai traffic, climate, camera and road-surface conditions.

VALIDATION SCORECARD

METRICS TO MEASURE

PRECISION **TP / (TP + FP)** How many flagged objects are correct?	RECALL **TP / (TP + FN)** How many real defects are found?
F1 SCORE **2PR / (P + R)** Balance of precision and recall.	mAP@0.5 **Mean AP at IoU = 0.50** Detection quality across classes.
LATENCY **ms/frame or FPS** Can inference keep up with capture?	LOCATION ERROR **Median + 95th percentile (m)** Are mapped detections near the road?
DUPLICATE RATE **Repeated reports / km / period** Are repeated sightings merged?	SERVICE TIMES **Median triage / repair time** Does workflow performance improve?

No model accuracy or field-impact values are claimed until measured on a documented holdout / pilot.

INTERFACE VERIFICATION PLAN

- Basemap loads; attribution visible
- Marker matches the selected record
- Search and severity/status chips work
- Sort order changes as requested
- 24h / 7d / 30d boundaries correct
- Popup opens the matching detail drawer
- Status actions refresh shared views
- Layout resizes on small screens

Evidence to retain: test ID, expected / actual result, pass or fail, browser version and dated screenshot. Server persistence and cross-user concurrency need separate tests.

SMART-CITY VALUE & LIMITATIONS

PRIMARY USERS

Municipal road engineers • maintenance coordinators • infrastructure analysts • repair crews

OPERATIONAL VALUE

- Shared spatial + record context
- Visible severity and open-work backlog
- Consistent triage and status review
- Foundation for segment-level hotspot analysis

SDG 11 ALIGNMENT: supports safer, more sustainable urban transport and integrated city planning. Outcomes require reliable data and maintenance follow-through.

LIMITATIONS: GPS quality, false alarms, image conditions, local domain shift and operator adoption can change results.

ROADMAP TO FIELD VALIDATION

01	CONNECT	Validated GPS schema + API ingestion
02	PERSIST	Database, user roles, audit trail, backups
03	BENCHMARK	Train / compare detector; test on local images
04	ANALYZE	Road snapping, spatiotemporal dedupe, hotspots
05	PILOT	Bounded ward trial; measure maintenance outcomes

SUCCESS MEASURES

Model: precision, recall, F1, mAP and latency. Operations: median GPS error, duplicate rate, time-to-triage, time-to-repair, reopened reports and reviewer agreement.

REFERENCES & CREDITS

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Map tiles: CARTO basemap. Map data credit: © OpenStreetMap contributors — openstreetmap.org/copyright.

PROJECT STATUS: working browser-based review interface; live ingestion, trained-model inference, persistent storage and measured field outcomes require further integration and validation.